

Phase 05 — Revision Sheet (Active Recall)

Do NOT open the notes. Answer aloud/on paper, then check and mark /. Re-test  items tomorrow.
Review: after phase → +1 day → +3 days → +1 week → +1 month.

A. Rapid-fire recall

1. What is NGINX? What was the C10K problem and how does NGINX's architecture solve it?
2. Forward proxy vs reverse proxy — who does each hide? Which is NGINX's main role here?
3. List 5 "front door" jobs NGINX does better than Spring Boot.
4. What is TLS termination? 3 reasons to terminate at NGINX instead of the app.
5. How does one NGINX serve many domains on one IP? (which header, which directive?)
6. The safe config-change workflow — 4 commands in order. Why `reload` over `restart`?
7. Recite a minimal reverse-proxy server block (`proxy_pass` + the 4 `proxy_set_header` lines).
8. Why do Host / X-Forwarded-For / X-Forwarded-Proto matter? What breaks without them?
9. 502 vs 504 behind NGINX — root cause of each, and the 3 checks for a 502.
10. Where are `access.log` and `error.log`, and what does each contain?
11. How does certbot/Let's Encrypt prove you own the domain? What does certbot change in your config?
12. Recite the HTTP→HTTPS redirect server block.
13. What do these do: HSTS header, X-Frame-Options, `server_tokens off`?
14. Rate limiting: what do `limit_req_zone rate=10r/s` and `burst=20` mean?
15. What causes a 413 error on uploads and the fix?
16. Static files: which location block pattern + headers make browsers cache assets 30 days?

B. Write from memory

- Full production server block: `api.shop.com`, HTTPS + redirect, proxy to `:8080`, forwarding headers, rate limit, gzip, 20MB uploads.
- The request path for `/product.png` vs `/api/products` through your config.

C. Command drill

```
nginx -t                systemctl reload nginx                tail -f /var/log/nginx/error.log
certbot --nginx -d site.com certbot renew --dry-run          ln -s sites-available/x sites-enabled/
```

D. Self-score

- 13+ of A → move on. 8-12 → redo lab + re-test tomorrow. <8 → re-read notes actively, redo lab.